

Locating biopiracy: geographically and culturally situated knowledges

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Abstract. The issue of ‘biopiracy’ inevitably evokes a wide variety of responses and emotions. It is likely that, for this reason, few academic articles have adequately or fully explored the discourse and/or issue. In this paper I analyse the strategic employment of the term and counterarguments generated by different parties including activists, nongovernmental organisations, academics, industry, and government representatives. Using the concept of ‘situated knowledges’ I explore how epistemic communities with intellectual property (IP) interests have generated one set of powerful and even hypocritical rhetorics and ‘harmonised’ them as global and universal truths. I then illustrate the political and cultural contingency of these IP discourses through a discussion of biopiracy cases and politics in Thailand, where IP laws have been rapidly and coercively imposed. Ultimately, I argue that the cultural, historical, and geographical specifics have been lost and obscured in an international debate that all too often employs ubiquitous understandings of ‘traditional knowledge’, ‘genetic resources’, and ‘protection.’

Introduction

In the environment created in the mid-1990s by the actions of transnational corporations, advancing biotechnologies, and international laws, nongovernmental organisation (NGO) campaigner Pat Mooney (2000, page 38) stated that “there is no ‘bioprospecting’ only ‘biopiracy’” (see also RAFL, 1995). Since that time, a considerable amount of debate has circled around the set of issues raised by these claims—issues of equity and justice for indigenous people, traditional healers, farming communities, and other groups staking claims over plants, animals and their component biological materials. In this paper I analyse the use (and denial) of the biopiracy discourse using Haraway’s (1991) concept of situated knowledges. Haraway’s ontological framework helps to break down some of the strategically employed and harsh dichotomies raised by Mooney and his contrarians for a vision of the intellectual property (IP) and biological resource issues that come from localised, partial, and situated perspectives. Specifically, I draw upon a number of biopiracy case studies in Thailand that highlight the cultural, geographical, historical, and political contingency of the now ‘harmonised’ global IP predicament. The regime now in place has demarcated the biological components of nature as ‘genetic resources’ (Whatmore, 2002, page 22) and all nonproprietary knowledge as the ‘public domain’ (Taubman, 2005, pages 543–544). Thus policy-maker assumptions about concepts such as ‘benefit sharing’ and ‘prior informed consent’ are also overgeneralised.

Haraway’s essay on the need to view knowledge as partial and ‘from somewhere’ stemmed from feminist studies critiques of universal objectivities:

“I am arguing for politics and epistemologies of location, positioning and situating, where partiality and not universality is the condition of being heard to make rational knowledge claims” (1991, page 195).

As Rose (1997, page 307) notes, this critique can be directed, not only at feminist knowledges, but also towards the material processes of power and knowledge that

maintain “scientific and technological, late industrial, militarized, racist and male dominant societies” (citing Haraway, 1991, page 188). The approach has been applied within frameworks for understanding the social construction (see Demeritt, 2002) or (in neo-Marxist terms) production of nature (Castree, 2001; McAfee, 2003) in the contexts of technological and economic manipulations of, and manifestations in, genetic resources. Further than that, actor-network theorists, such as Strathern (1999; 2004) and particularly Whatmore (1997; 2002), invoke an ontology of ‘hybridity’ to raise ethical questions about living with nature and each other, with an expanded sense of morally significant others, in a more than human world (Demeritt, 2005, page 818). Expanding upon this literature, I apply Haraway’s argument to the debate about IP, biodiversity, and associated knowledges, because of the dominance of IP discourses as if they are without origin and should apply universally. I argue instead that IP discourses—like the biopiracy discourse—are historically, geographically, and culturally contingent. While I utilise the language of actor-network theory sparingly, I raise similar ethical questions about nature in the context of ‘traditional knowledge’ that arise from the culture-specific embodiment and personification of plants by specific groups.

Research insights

Key insights for this paper were drawn from research involving sixty-four interviews of key stakeholders, biopiracy case studies (including the analysis of patent and IP documents), and local community customary law case studies in Thailand. Interviews engaged with a diverse group of individuals including officials and country delegates to the World Trade Organization (WTO) and World Intellectual Property Organization (WIPO) in Geneva, Switzerland, and key actors in Thailand (government officials, industry, academics, NGO staff, activists, researchers, farmers and plant breeders, and people from local ethnic minority groups) in 2005 and 2006. The research also involved observation of and involvement in a number of IP negotiations and high-level meetings. These occurred whilst working for and consulting to the International Centre for Trade and Sustainable Development in conjunction with the United Nations Conference on Trade and Development in Geneva, and for the United Nations Development Programme between 2005 and 2008.

The discursive politics of biopiracy

‘Biopiracy’ is a term that was first coined by the Canada-based NGO Rural Advancement Foundation International (RAFI) (now the ETC Group) activist Mooney (2000; RAFI, 1995). The term was developed because of growing frustrations about the appropriation and monopolisation of long-held medicinal and agricultural knowledge about nature, as well as the related physical resources (plants, animals, and their components). The flow of these resources and knowledge, often “from biodiversity in the South to medicines, [cosmetics and crops] in the North” (Svarstad and Dhillion, 2000, page 9), was targeted by NGOs as a hypocritical injustice on the part of corporations and researchers predominantly from the US, Europe, Japan, and the Western world. Although the colonial enterprise of plant and animal collecting has been ongoing for centuries (see, for example, Brockway, 1979; Kloppenberg, 1988; Schiebinger, 2004), the biopiracy discourse was generated to illustrate that more recent technological and institutional changes have encouraged new inequities. In the context of the new ‘global’ IP rules, biopiracy has essentially been wielded as a counterdiscourse to IP ‘piracy’.

As Mooney (2000) notes, international agreements such as the United Nations Convention on Biological Diversity (CBD) negotiated at the Rio Earth Summit in 1992, and the WTO Agreement on Trade-Related Aspects of Intellectual Property

Rights (TRIPS) consolidated some specific Euro-American ideologies about the IP ownership of nature as genetic resources. Specifically, the term ‘bioprospecting’ was coined in response to a book by the World Resources Institute (Reid et al, 1993) entitled *Biodiversity Prospecting*. The book championed a ‘win–win’ scenario of ‘benefit sharing’ and respect for indigenous or traditional knowledge, and also a boon for humanity through scientific research. Mooney and RAFI disagreed with these assumptions, using biopiracy to point out that the socioeconomic environment at the time made mutually beneficial contracts between providers and users of genetic resources virtually impossible. According to Mooney (2000), intergovernmental regulatory mechanisms and agreements were needed for the possibility of equitable benefit sharing. But, further than that, their campaign targeted the explicit protectionism that biotechnology companies were implementing through not only IP rights, but also biotechnological advances such as genetic-use restriction technologies—which they dubbed “terminator technologies” (RAFI, 1998). RAFI linked biopiracy claims with evidence that transnational seed companies were planning to use terminator technologies for seed sterilisation, thus inhibiting farmer reuse of seeds. In this way, RAFI was able to highlight the ongoing legal and technical institutionalisation of farmers from the global South.

Biopiracy was picked up and used by many other activists and NGOs for related purposes and campaigns. Probably the most prominent individual champion of the biopiracy discourse is Shiva (2001):

“Biopiracy refers to the use of intellectual property systems to legitimize the exclusive ownership and control over biological resources and biological products and processes that have been used over centuries in non-industrialized cultures” (page 49).

Shiva’s attacks on the green revolution (1991), the IP system (1997; 2001), and genetic reductionism also gained significant international attention. Particularly relevant was her activism on several basmati rice patents in the US, which led to the revocation of four basmati-related claims by the patent owner (NISCAIR, 2002). Behind Shiva was a network of farmers and local communities with sufficient power to influence public opinion and government policies, laws (such as The Protection of Plant Varieties and Farmers’ Rights Act 53 of 2001 in India), and positions in international fora. This network of influence was expanded through engagement with other NGOs such as the Third World Network, in statements by indigenous people’s movements and civil society actions such as the Thammasaat Resolution in Thailand (in 1997), and also in the discourses of intergovernmental organisations of developing countries such as the South Centre.

Thus, with the enrolment of broader public sympathies for NGO causes, discussion on biopiracy has extended beyond ‘activist’ agendas. What are particularly interesting are the recent discussions on biopiracy by government delegates in statements and submissions to international organisations:

“One of the measures adopted [in Peru] was the creation of the National Anti-Biopiracy Commission, whose basic task is to develop actions to identify, prevent and avoid acts of biopiracy which involve biological resources of Peruvian origin and traditional knowledge of the indigenous peoples of Peru” (Peru, 2005, page 1).

“‘There are lots of problems with biopiracy in Thailand ...’ despite the introduction of national laws, showing the need for international regulation” (*IP-Watch* 2006).

With the development of national bodies targeting biopiracy and official statements by government departments and delegates to the WTO, attempts to mitigate biopiracy have gained increasing momentum. Official engagement with biopiracy implies public

pressures on governments to find equitable and just solutions to the issues of biological resource and knowledge ownership. But it is also based on the pressures within some states to protect (and exploit) their 'sovereign rights' to biodiversity according to the CBD. Out of this convention, the discourses of benefit sharing and prior informed consent are spun to serve variable interests from NGOs like RAFA, to government departments, and biotechnology industry lobbies (see Finston, 2005).

In addition, biopiracy has been used strategically in negotiations. Owing to the inequities in trade flows between countries in the global South and those in the global North, biological resources have emerged as a bargaining tool for those countries with less political and economic leverage. For example, Dutfield (2005) notes that discussions on a 'disclosure of origin requirement' to limit inappropriate patenting of acquired biological resources have been used strategically by developing countries to stall negotiating progress on a Substantive Patent Law Treaty in WIPO.⁽¹⁾

These governmental resistances also hint at a growing awareness of the new IP 'fundamentalism' (Dutfield, 2006) and 'inequality' (Drahos and Braithwaite, 2003), as raised by influential academics and researchers, which is best illustrated through a historical analysis of the situated and hypocritical policies of countries like the US and Japan, and those from the EU.

Situated knowledges of intellectual property

Several academics have documented the recent history of the IP system's expansion (Braithwaite and Drahos, 2000; Drahos and Braithwaite, 2003; Dutfield and Suthersanen, 2005; Sell, 2003). Drahos and Braithwaite (2003, page 10) note that, during the course of an interview in 1994 with a senior US trade negotiator, he remarked that "probably less than 50 people were responsible for TRIPS" IP lobby groups representing film, software, and pharmaceutical industries, such as the International Intellectual Property Alliance, were responsible for the creation of the discourses of piracy (as used in the context of copyright infringement) and 'trade-related' IP.⁽²⁾ As Drahos and Braithwaite (2003, pages 19–38) note, historically piracy had nothing to do with copying of copyright products, and copying was not 'stealing' (Dutfield, 2006). Rather, piracy was recreated and used to leverage support from governments, industries, and the public for the protection of IP rights across the globe. In fact, Dutfield and Suthersanen (2005, pages 142–143) note that the biggest current proponent of IP expansion—the US—had a particularly negligent record for foreign copyright protection during the 19th and 20th centuries as its industries developed and flourished, and it did not sign the Berne Convention for the Protection of Literary and Artistic Works until 1989. In this light, the current approach of the US to IP expansion and enforcement seems highly hypocritical.

In the lead-up to the Uruguay round of Multilateral Trade Negotiations, which ultimately created the WTO in 1994, the US and some key IP allies began enrolling developing countries through bilateral and regional trade negotiations, the threat of sanctions through the US's 'Special 301' actions, trade leverage, and 'green room' meetings (Jawara and Kwa, 2003). By the time the negotiations were complete, most

⁽¹⁾ This was also implied in a number of confidential interviews and meeting comments with developing country delegates to the WTO and WIPO from regions such as South Asia, South America, and Africa. Their dissatisfaction with progress on 'development agenda' items and the terms of trade more generally were impetus for such a strategy.

⁽²⁾ A small group of consultants, lobbyists, and lawyers (notably Eric Smith, Elery Simon, and Jacques Gorlin) developed the discursive linkage of IP with trade, which fed through the Advisory Committee on Trade Negotiations to the US Trade Representative (USTR) (Braithwaite and Drahos, 2000, pages 61–62).

countries had conceded to the concept of ‘trade-related’ IP (Braithwaite and Drahos, 2000, pages 179–181). In the words of Haraway, the lobbyists, with the help of the US trade representative and their supporters (the EU, Japan, and Canada), were able to perform a ‘god trick’ and redefine IP as placeless, universal, and global (Wright, 2005). What is even more worrying is the ongoing trend of bilateral and regional ‘free-trade agreements’ that continue to raise IP standards of protection in a range of areas, which Drahos (2001; 2003) calls the “TRIPS-plus ratchet”. These bilaterals can be used to undermine the progress in multilateral discussions on important ‘development agenda’ issues such as access to medicines, access to affordable educational materials, and patents on life forms or plant variety protections (see Meléndez-Ortiz and Roffe, 2009).

Not long after TRIPS was signed, developing country delegates were seeking concessions to some of the IP minimum standards that it required. In relation to the IP protection of biological resources, required by Article 27.3(b) of TRIPS, the African Group (2003) made the following submission:

“Patents on life forms are unethical and the TRIPS Agreement should prohibit them Such patents are contrary to the moral and cultural norms of many societies in Members of the WTO. They make the exception ... for protecting ordre public and morality, which Members that consider patents on life forms to be contrary to the fabric of their society and culture, and to be immoral, and which they would otherwise invoke, meaningless in this regard” (page 2).

The inclusion of patent (or other IP) protection requirements on life forms clearly neglects to adequately consider the moral and cultural norms of many societies, to both the technologies involved and the monopolisation of life-forms. Yet, for the advancement of the interests of agrochemical cartels, these were coercively forced into the text of TRIPS (see Dutfield, 2003, pages 195–204). As Braithwaite and Drahos (2000, page 75) note, the IP community operating behind these actions was an epistemic community “comprised of transnational elites with important intellectual property portfolios to protect—and their lawyers.” Their solution, as Dutfield (2003, page 198) notes, was to effectively impose on the world

“their interpretation of fair competition in high-technology ... by means of the global standardization of national IP regulation as far as possible equivalent to the standards existing in the US.”

Thus, a newborn culture of specific economic protectionism trumped a plethora of long-held and widely dispersed moral and cultural norms about the treatment of nature, such as those expressed by the African Group, those in the Indigenous Peoples Earth Charter⁽³⁾ (also see Ingold, 1989; Posey, 1999), and those that I illustrate in relation to particular regions, cultural groups, and biopiracy issues in Thailand.

Counter-counterdiscourses against biopiracy

A number of authors have refuted the existence of biopiracy by carefully selecting some claims by NGOs, highlighting their vagaries and essentially arguing the lack of a legal basis for biopiracy (Chen, 2006; Heald, 2003; Oxley, 2005). Their arguments are entirely based on economic (utilitarian) and legal grounds, and are thus also partial and situated, despite their claims for universal dismantling of the ‘myth’ of biopiracy (Chen, 2006, page 6):

“Ethnobiological knowledge is best characterised as the inventive transformation of genetic information into commercially valuable applications. An appropriately utilitarian view of property and its relationship to each layer of biological information thus dissolves any allegation of biopiracy.”

⁽³⁾ *Kari-Oca Declaration and the Indigenous Peoples’ Earth Charter*, 1992. World Conference of Indigenous Peoples on Territory, Environment and Development, Kari-Oca, Brazil, 30 May 1992.

Who is Chen to judge how ethnobiological knowledge is 'best' characterised? Who is a utilitarian view of property and biological information 'appropriate' to? These characterisations might seem appropriate to some IP lawyers, some economists, and some biotechnologists, but they are probably not the best characterisations or appropriate assumptions for people who view the world from a multiplicity of other perspectives. The protagonists behind IP expansionist agendas and the biopiracy counterdiscourses present only a narrow worldview, yet their success has been considerable, through the translational enrolment of key actors (Latour, 1986) backed up by trade leverage tactics.

The counterbiopiracy authors also make overextended assumptions about what proponents of a 'disclosure requirement' and the biopiracy discourse have been seeking. Oxley (2005, page 4) suggests that amendments to TRIPS and the CBD rules are being sought "so developing countries can reap the benefits of their natural biodiversity—'green gold'." Similarly, Chen (2006, page 29) suggests that

"The real point of the biopiracy narrative is that the global south wants its largest possible share of the world's wealth."

These assertions are misguided and overly narrow. As I have noted already, biopiracy is also a counterdiscourse to hinder the historically hypocritical ratchet of IP norms and standards. It is also a reaction by a diverse range of different cultures and perspectives to IP monopolisation and the genetic reductionism of nature—these agendas go well beyond 'rent seeking'.

For example, the University of Hawai'i has decided to relinquish claims to three patents on taro plant varieties in recognition of its cultural significance to indigenous Hawai'ians. Taro is considered a sacred ancestor of the Hawai'ian people. According to Sarah Sullivan of Hawaii Seed, "culturally significant plants such as taro should not be owned", and Molokai activist Walter Ritte Jr, "you can't change our ancestors without our permission" (ICTSD, 2006). The cultural embodiment and personification of plants has been largely ignored by the deniers of biopiracy. Obviously, not all plants will have the same degree of cultural significance amongst indigenous or local groups, but there are often complex culturally or spiritually informed sentiments about ownership and genetic modification of plants and animals (Ingold, 1989; Posey, 1999). In the case of patents, ownership extends to all specimens and products of the patented invention. So, in this case the patents covered all cross-bred leaf-blight-resistant taro plants. The fact that the variety has been altered is significant for the criterion of invention required in a patent. In the cultural embodiment context of the Hawai'ian people, it can be assumed that, altered or not, they would not want a form of private monopoly ownership over their ancestors.

Most of the biopiracy counterdiscourses do not 'dismantle' or 'dissolve' biopiracy—they merely reconfigure it to their own situated knowledges. The expansion of multilateral negotiations on a disclosure of origin patent requirement in the TRIPS Council and WIPO, and an International Regime on Benefit Sharing in the CBD, highlight the momentum of 'reformist agendas' critical of the erroneous patents, unfair and inequitable access to biological resources that have been associated with biopiracy. Yet these negotiations are underpinned by universalising ideologies and harmonising (ie globalising) objectives that see ubiquitous use of terminologies such as 'genetic' or 'biological' resources, traditional knowledge, and 'folklore' (see Strathern, 2004). The amendment of these international regimes will replace one harmonised set of rules with another, with the ultimate cost being to the diversity of culturally informed perspectives on nature—particularly life-forms. In order to illustrate some of these diverse local knowledges and perspectives, I explore some of the biopiracy cases and concerns with reference to local accounts and interviews from Thailand.

Thai contexts and case studies

The introduction of stricter IP conditions in Thailand has received widely varied responses from the public, interest groups, and stakeholders. Thailand had been on the US Special 301 watch list for a number of years under threat of trade sanctions and removal of US trade preferencing under the WTO Generalised System of Preferences (Kuanpoth, 2004). These pressures, along with requirements according to ratification of the WTO agreements, saw Thailand develop new laws and an International Trade and Intellectual Property Court (Ariyanuntaka, 1999). Copyright has existed in Thailand only for a century and IP as a concept has a history of only three decades, since the drafting of the first patent act.

In the case of biodiversity and traditional knowledge, significant pressures from huge farmer's networks such as the Alternative Agriculture Network, the Assembly of the Poor, the Community Forests Network, and NGOs such as BioThai pressured the government to act to prevent biopiracy. Academics (including 'sociological' IP experts Jakkrit Kuanpoth and Jaroen Compeerapaap) and prominent government officials such as Professor Saneh Chamarik (Chairman of the National Human Rights Commission of Thailand) and Dr Pennapa Subcharoen (a senior official in the Department of Public Health) also lobbied for the creation of laws that sought to respond to these biopiracy concerns. The laws developed: The Act on the Promotion and Protection of Thai Traditional Medicinal Intelligence (1999, BE 2542), and the Plant Variety Protection Act (1999, BE 2542), were truly *sui generis* in that they sought to go beyond international norms and incorporate specific local protection and promotion systems (Robinson, 2007). The government, influenced by prevalent culturally informed perspectives and pressure from these groups, also opted to exclude plants, animals, and naturally occurring microorganisms from patentability under the Thai Patents Act (1979, as amended 1992 and 1999, BE 2522) (Assawin Tharang Kung, interview, 2005). This same network pushed for delays in the ratification of the CBD until 2004, particularly over concerns about the wording of "facilitated access to genetic resources" (Compeerapaap, interview, 2005). At the same time there have been conflicting pressures within government as influenced by biotechnology science and industry lobbies, for the facilitation of access to Thai genetic resources and the development of associated industries.

There are a complex array of catalysts which spurred considerable public sympathy for biopiracy complaints in Thailand and the perceived need to protect biological resources and traditional knowledge (see Baker and Pongpaichit, 2002; Lohmann, 1991). Briefly, land clearing and deforestation in the past century have been so extensive that, when Thailand joined the CBD, the Megadiverse group of countries refused to include Thailand as a negotiating member (Srethasirote, interview, 2005; Compeerapaap, interview, 2005). The remaining tracts of forest, particularly in the north and west of the country, have become the subject of intense debate in recent decades. This was largely focused between local minority communities who had established their communities in forest areas (with the support of a number of academics and NGOs) and opposing parties that had adopted more exclusionary ideologies about biodiversity conservation (government forestry and agriculture departments, and 'conservationist' NGOs). The importance of local traditional knowledge was one of the key arguments put forward by the community forests network (Ganjanapan, 2000; Santasombat, 2003). Yet government documents such as CBD national reports barely acknowledge local in situ folk knowledge, particularly that of minority groups, often focusing instead on more formalised and documented 'national' Thai traditional knowledge.

Changing technological conditions under the green revolution and through biotechnologies have created divisions between 'traditionalists' (including those influenced

by royal discourses on 'agricultural heritage') such as those from the Alternative Agriculture Network who support the cultivation of noncommercial crops, and the proponents of 'improved' seeds.⁽⁴⁾ Although Thailand's biotechnology industry is in its infancy, the influence of seed and crop multinationals such as Monsanto, DuPont Pioneer Hybrid, and Syngenta is omnipresent. This is evidenced by the rapid implementation of 'new' plant variety protections under Thai law, while 'general domestic', 'wild', and 'local' plant variety protections are still yet to be implemented (Siripat, interviews, 2005; Srethasirote, interviews, 2005–06).

Biopiracy became embroiled amidst these preexisting and evolving politics in the late 1990s. The most widely publicised cases (see, for example, Kuanpoth, 2005; Robinson, 2006; Tanasugarn, 1998) of biopiracy or 'misappropriation' have been those of jasmine rice or *khao hom mali* for which there are a number of incidents to consider. Why have these been so politicised? It is worth noting that rice represents Thailand's largest agricultural export for which *khao hom mali* makes up a significant proportion. Furthermore Thailand's long history as a rice-growing country whereby rice (*khao*, literally, 'food') represents an integral part of its agricultural heritage, emphasise its importance both economically and culturally. Therefore, the following two cases hold a place of considerable concern for the Thai people, and for Thai authorities. The first case examined is the Jasmati case.

Jasmati case

In 1998 the Thai NGO BioThai was informed that a US company had registered jasmine rice under the US IP system. Scrutiny of US Patent and Trademark Office documents reveals that a US company named Ricetec, Inc., had succeeded in registering a trademark associated with a rice product called 'Jasmati' in 1993.

The concerns were rapidly and widely reported in the Thai press with varying degrees of accuracy.⁽⁵⁾ Because Ricetec had also successfully registered a US patent called "Basmati rice lines and grains" in 1998, some confusion may have occurred between the two cases (Rerkasem, 2003). It was widely believed by the Thai public at the time that there was a trademark *and* a patent on jasmine rice germplasm. In fact there was only the Jasmati trademark (Tanasugarn, 1998).

Consequently, there were a number of protests held throughout the country, including a crowd of about 500 jasmine rice farmers outside the US embassy in Bangkok led by the Assembly of the Poor and the NGO BioThai. These protests demanded revocation of the trademark and the cessation of pressures on developing countries to provide patent protection of life forms.⁽⁶⁾

The Jasmati trademark has since been approved for renewal in 2003. The Thai government and Thai companies to date have not challenged the trademark. A representative from the Thai Department of Intellectual Property (DIP) was somewhat sceptical whether a case for repeal of the trademark would succeed, and noted the expense for which a trial would amount (Assawin Tharang Gung, Interview, 2005).

Another relevant US patent has been identified with patent number 5208 063 assigned to Ricetec—this time the subject of a process that yields fluffy and tender

⁽⁴⁾ This is part of a broader traditionalist versus globaliser debate on public policy in Thailand (Reynolds, 2001) that has more broadly seen authors questioning the politics of urban versus rural environmental consciousness (Forsyth, 2004).

⁽⁵⁾ For example, the Thai-language *Matichon Weekly* (1998) placed an article on 23 July entitled "Farmers mob demand the American Embassy to revoke the Jasmati patent", despite there being no patent on the Jasmati product. The concern and confusion was probably related to the patent on Basmati by the same company.

⁽⁶⁾ Monsanto has also submitted letters to the USTR seeking to pressure Thailand to modify laws relating to biotechnology (Monsanto Corporation, 8 April 2004, available at <http://www.grain.org>).

cooked rice with similar characteristics to that of jasmine rice (Tanasugarn, 1998).⁽⁷⁾ This patent arouses curiosity about whether Ricetec were intending to use this process to give the Jasmati product characteristics more like Thai jasmine rice, thus contributing to possible deception regarding the name and origins of the product.

The case highlights fairly widespread Thai public concerns that plants and animals should be excluded from monopolistic IP protection (see Changtavorn, interview, 2005; Lianchamroon, interview, 2005; Yuthavong, 2003). As Tanasugarn (1998), a Thai Supreme Court judge, argues, many Americans who complained that the jasmine rice crisis was blown out of proportion to its actual severity should understand that the Thais' resentment towards any attempt to monopolise plants is based on their agricultural background and heritage. Since rice is the most important crop culturally and economically for Thailand, the issue was particularly sensitive. As Boonridrerthaikul (2004, page 28) notes:

"Rice for Thai people has profound significance not only as a source of food but also as an essential component of religious practice, national culture, and way of life. Their thinking with respect to rice is demonstrated in their beliefs, rituals, traditions, folklore, proverbs, riddles, folk songs which are evident throughout production and even consumption processes."

Nathalang (2004), for example, describes the significance of rice spirit (*khwan*) through Thai folklore in the tale of *Mae Phoesop*—the spirit protector or provider of rice. She indicates that without rice a person cannot live long, and that Thai people should be respectful of Mae Phoesop at mealtimes. Nathalang (2004) also describes rituals invoking the spirit in order to return plentiful crops. These beliefs (derived from a fusion of Buddhist, animist, and folkloric teachings) are still popularly followed and can be readily witnessed from virtually any rural highway in Thailand, where spirit houses are set up on individual properties to protect their crops.

Stepwise Programme for the Improvement of Jasmine Rice for the US

The 'Stepwise Programme for the Improvement of Jasmine Rice for the US' has also been the subject of biopiracy (of tangible and intangible property) as well as competitiveness concerns. The programme has been the subject of debate in numerous meetings in Thailand and has appeared in the press with outraged comments made by NGOs and members of the Thai public (see *The Nation* 2001).

The purpose of the programme is to develop jasmine rice that US farmers can grow easily and profitably, and thus compete with Thai imports according to the Dale Bumpers National Rice Research Center, Arkansas. Public concern and news media coverage in Thailand has focused primarily on the partner project run by plant breeder Chris Deren in the University of Florida Everglades Research and Education Center (see, for example, *Bangkok Post* 2003; *The Nation* 2001). The programme uses two methods including induced mutations of jasmine rice using gamma-ray irradiation, and using conventional breeding methods.

There are some moral, cultural, or religious concerns to raise here. Inducing mutation through gamma-ray bombardment would commonly be perceived as being at odds with popular Theravada Buddhist teachings and 'Thai values' of nature that are embedded in folklore and daily mealtime rituals or practices described by a number of Thai academics [Kriengkraipetch (1989), Laohavanich (1989); and Chamarik, interview, 2005, respectively].

Regarding the conventional method, Deren has been cross-breeding the Thai rice strain khao kao dawk mali 105 with Jasmine 85. Scientists selected Jasmine 85 from the

⁽⁷⁾ US Patent and Trademark Office (USPTO) patent 5 208 063, "Process that yields fluffy and tender cooked rice" (<http://www.uspto.gov>).

International Rice Research Institute (IRRI). It is likely that these strains of rice were collected in 1950, by Mr Sunthorn Srihaneon, an agricultural official in Chachoengsao province. The strains were grown locally in Chachoengsao and Chonburi. The collection of various rice strains was part of a project operated with an American geneticist, Dr Henry H Love, acting as consultant. The qualities of the rice were investigated and then later submitted to IRRI (Lianchamroon, interview, 2005; Srethasirrote, interview, 2005). Thus the Stepwise Programme uses two varieties that originated in Thailand and are now being modified to suit US conditions.

One main concern is that Thai farmers in the arid *Thung Kula Ronghai* plateau in the Northeast (*Issan* region) have through generations of conventional breeding and selection improved the quality of jasmine rice from its likely origins in Chachoengsao. *The Nation* newspaper reports that these Issan farmers had never thought of seeking exclusive rights to the fruit of their labour. They freely shared seeds with other farmers. To them, and indeed many other farmers in the region, rice and other food crops are the 'common heritage of (hu)mankind'—a principle now essentially being eroded by the patent system (*The Nation* 2001).

The other main concern regarding this 'misappropriation' is that US breeders can reduce the need for the US to import jasmine rice products from Thailand, and also compete with Thai jasmine rice in international markets. When the germplasm was transferred, the Thai researchers did not perceive that it would be used to compete against the exports of Thai farmers (Suntorn Srihaneon, meeting comments, 2005). These concerns have been expressed by the rice industry, the Ministry of Commerce, and also local farmers.

In addition, concerns about the potential for a US patent on the rice have been expressed. Chris Deren's rice is very similar to a variety of Thai jasmine rice called *khao pathumthani 1*, meaning that there could be customs restrictions or delays while grain entering the US is examined. Patented or not, Deren's varieties could also have consequences for Thai exports. The US-bred varieties could essentially 'free ride' on the donor germplasm to provide an increased market share in the US rice industry, thus reducing demand for Thai exports.

On 6 November 2001 IRRI issued a press statement in response to concerns in the Thai press. IRRI admitted that in 1995 they shared a sample of Jasmine 85 with the Dale Bumpers Rice Research Center—however, no material transfer agreement (MTA) was signed because at that stage they were not required by IRRI. Both scientists have since publicly agreed to accept all the terms and conditions in the current IRRI MTA and have issued signed letters saying so (IRRI, 2001).

Thai and US patents on *kwao krua* (*Pueraria mifrica*)

Considerable confusion has also surrounded a number of attempts to patent compositions containing *kwao krua* (*Pueraria mirifica*). The herb *kwao krua* has been known for its cosmetic and revitalising qualities for more than 100 years by Thai healers, communities, and households. In more recent years scientists have identified that the effects are related to the presence of phyto-oestrogens. Scientific claims have since been made that the extracts may enlarge and firm breasts and assist with male sexual performance.

A number of patents have been filed in Thailand which cover extracts or compositions containing *kwao krua*. Some of these have reportedly been used maliciously, through the inflation of patent scope, to exclude other herbalists from the sale of their products (see Tanasugarn, 1999). Under the Patent Act of Thailand (1979, as amended 1999) these patents were allowed on the basis that they were a *chemical derivative* of a plant product as part of a composition. Thus they narrowly avoided the scope of the

patent exclusion on plant and plant extracts under Section 9(1) of the Patent Act (Assawin Tharang Gung, interviews, 2005 and 2006). There are therefore concerns from Thai IP experts and health officials that local patents on kwao krua extracts may inhibit their commercial sale by other local businesses (Subcharoen et al, 2001).

Thai scriptures collected since 1931 by Luang Anusarn-Sunthorn provide evidence of the prior art of kwao krua in considerable detail. Clearly documented, the knowledge of kwao krua, is national public domain knowledge with an extensive history, but is also a form of traditional folk medicinal knowledge (the herb is used by communities in the Issan region). In the examination process the DIP did not know about the documents establishing prior art. Had they known, the claimed invention would likely have failed a nonobviousness test. Officials at the DIP are now waiting for someone to bring the case to court (Assawin Tharang Kung, interview, 2006; Robinson and Kuanpoth, 2009; Tanasugarn, 1999).

Following this first case, a number of subsequent Thai and US patents have also emerged. A Thai scientist is the claimed inventor of three different inventions under patent examination in DIP Thailand relating to kwao krua. These apply to Thai patent numbers 046 779, 048 605, and 052 443, of which the last has been the source of considerable controversy. According to the head of the Patent Division of the DIP, the approval of these patents has been challenged, and has taken 2–3 years in the Thai Intellectual Property and International Trade Court, and then 3–4 years in the Supreme Court (Assawin Tharang Kung, interview, 2006). Each of the patents refers to extracts of one or more of *Pueraria mirifica* (white kwao krua), *Butea superba* (red kwao krua), and/or *Mucuna collettii* (black kwao krua) (Robinson and Kuanpoth, 2009).

As attorney and academic Dr Jade Donovanik (*Bangkok Post* 2004) has described, Thai patent number 052 443 appears to be the direct foundation of USPTO patent number 6 673 377 (interview, 2005; IPIC Thailand, 1998). The claims, in both the Thai and US patent applications, include an extract of kwao krua, a method for extraction, and a method for manufacture. While the patents on processes and methods seem valid, the primary concern here is that the *extract* is not particularly new, inventive, or nonobvious.

Korean company Cheil Jedang Corporation holds US patent number 6 673 377 for a product which uses white kwao krua as its main ingredient.⁽⁸⁾ The basis of the Korean patent is:

“an extract derived from *Pueraria mirifica* having an effect on improving breast firmness, breast enlargement and wrinkle removal from the breast, wherein said extract is prepared by the steps of: drying tubers, roots, stems, leaves and/or tissue-cultured calluses of *Pueraria mirifica* ... extracting the mixture; and filtering the resulting extract and then concentrating it in a vacuum to remove the solvent.”

The details in the three Thai patent applications and the abovementioned US patent bear striking similarities to entries in Luang Anusarn-Sunthorn's scriptures. This has outraged many Thai healers, NGOs, and even government officials from the Department of Public Health. There is a primary concern that the patents may inhibit exports to the US of similar products which contain kwao krua extracts. This could occur through litigation and customs restrictions. There are also serious concerns about excessive collection within Thailand of kwao krua, which is a slow-growing vine,

⁽⁸⁾ USPTO patent number 6 673 377, 6 January 2004, “Extracts derived from *Pueraria mirifica*, *Butea superba* and/or *Mucuna Collettii* and extraction thereof” (<http://www.uspto.gov>). There is also another US patent, 6 352 685 “External preparation for the skin”, which has accrued similar biopiracy claims and concerns (<http://www.uspto.gov>).

due to the increased publicity it has received since the patents were granted (*Bangkok Post* 2004; Subcharoen et al, 2001).

Donavanik (*Bangkok Post* 2004) has made a persuasive argument that these claimed inventions under patent have little innovative merit. Arguably, the documented public domain traditional knowledge should have been detected on the basis of standard patent criteria. This is something that the DIP have expressed concerns about, and the case has been one of the main bases for attempting to establish traditional knowledge databases or registers for utilisation by patent examiners in Thailand (Assawin Tharang Gung, interview, 2006; Donavanik, interview, 2006). But because the US does not recognise foreign prior art as a criterion for disallowing a patent, such an approach would have limited impact extraterritorially.

It should be noted that the plant is endemic to the region, not just Thailand. The plant exists in Burma and Laos (and possibly also Cambodia) (*The Nation* 2005). Knowledge of the traditional uses of the plant, through documentation and public use, has entered the public domain. Yet it has been claimed by some healers that the knowledge has origins in the Issan region where the plant is most prevalent (Patthi Dang, interview, 2006; Patthi Ta Yae, interview, 2006). This poses important questions about prior informed consent—who should be able to authorise it? If consent was authorised, how would ‘fair and equitable benefits’ be provided to custodians of the plant and associated traditional knowledge? One possible solution could be to provide benefits to traditional healers’ networks which conserve the herb, if profits were created from the commercialisation of patented kwao krua derivatives. But it is not clear if many local groups are actually seeking anything as superficial as benefit sharing. As discussed in the last section, objections to the appropriation, commercialisation, and IP protection of plants and their associated knowledge are more complex and geographically and culturally specific than the CBD rules and the biopiracy contrarians assume.

Diverse responses

In her study, de Laet (2000, page 149) indicates that

“patents are different things in different places—and it is their travel that makes them so ... the patent [i]s both a changeable object and an agent of change.”

The same can be said of other forms of IP (trademarks and plant variety protections) and the local nature of nature (or biodiversity)—there are place-specific and culture-specific contexts in which they are envisioned, where they act or operate in different ways. In this final section I draw on some ethnographic observations, interviews, and surveys which demonstrate the diverse range of specific responses to issues raised by biopiracy and IP, based on different cultural and geographical locations within Thailand.

Academic and policy-maker survey

With the development of the TRIPS agreement, the often conflicting objectives of the CBD have been skewed by an expanded proprietarian consciousness and by proprietary interests. As a result, the CBD objectives referring to the equitable sharing of benefits arising from the use of genetic resources may be interpreted as Whatmore (2002, page 22) describes, such that

“[t]he logic here is to establish a universal exchange rate between the scientific value of animal (and plant) life, measured in units of biological rarity, and that most pervasive of human currencies—economic value.”

A range of other authors have also made similar complaints about the universalist reduction and revaluation of nature (McAfee, 1999; 2003; Castree, 2003; Parry, 2002)

and of knowledge (Haraway, 1991). Thus, the CBD provides a tense forum for negotiating its different interpretations relating to the protection of traditional knowledge of those that conserve biodiversity, and the often conflicting pressures for facilitated access and exploitation of genetic resources. On (biodiversity-related) “traditional knowledge and folklore”, a survey of a group of twenty-five key academics, NGOs, and government officials provide some insights into comparable sentiments from perspectives in Thailand.

In the survey, one question asked “What should traditional knowledge and folklore protection be aimed at?” The largest response was “to prevent people from inappropriately taking advantage of traditional knowledge and folklore”, followed by the “restoration and promotion of traditional knowledge and folklore”, and the “preservation of traditional knowledge and folklore for broader social benefits”. Economic reasons factored in as a lesser response. These responses seem to reject the revaluation of knowledge and also reflect some concerns in local case studies, that *protection* may have little to do with economic incentives or external ‘benefits.’⁽⁹⁾

Local contexts from Karen communities

The *Khon pga k'nyau* (Karen) people from two communities (Baan Mae Ka Pu and Baan Soplan, Amphoe Samoeng, Chiang Mai Province) expressed a number of concerns surrounding the exploitation of nature, of knowledge, and of biopiracy. One elder in Soplan, Pathii Taa Ye (interview, 14 February 2006), indicated that, when their medicinal knowledge was useful, where it could help other people, he was happy to share it with them. Sharing was a typical community principle and lifestyle for their people. He then complained, however, that under the system of IP he no longer knew what could happen to such knowledge if he shared it with outsiders. He had become sceptical. Pathii Taa Ye knew about the potential misappropriation and/or IP protection of biological materials, but did not know what that meant for them. He was alarmed about the modification of plants and exclusionary controls over them, having detrimental effects on the *khwan* of the plants and on traditional healers.

Despite this apparent openness, in the culture of the *Khon pga k'nyau*, there are many customary norms, rules and, taboos about sharing such knowledge or resources with outsiders. Notably:

“Medicines and herbs should not be hoarded. They should only be collected as needed for the treatment of sick people. Herbs should not be overcollected, especially if the plant is rare. Before going to collect the herbs, the spirits of the herbs should be first asked and a small donation should be made to the spirits at a shrine” (Pathii Taa Ye, interview, 13 February 2006).

They also indicated that:

“When knowledge of herbs is shared it is often informed by their perception of the character of the person seeking the knowledge as well as based on possible indications from spirits about them. If the knowledge that is shared is used inappropriately, it may adversely affect the spirit of the person who provided the knowledge, in the form of illness or otherwise” (Pathii Dang, interview, 14 February 2006).

These customary rules highlight the failings of the positivist logic of Western liberal legal theory. These norms extend the ‘legal’ or normative jurisdictions of scale beyond that of local, national, regional, or global, to the internal or utterly intimate and to the infinites of life, death, and reincarnation (related to karma). These examples

⁽⁹⁾ Of the farmers’ groups I spoke to there were a few exceptions—those in Suphan Buri and some farmers in Ku Ka Singha, Roi Et, indicated that they would be interested in accruing ‘benefits’ *if they were likely*. However, the farmers and NGO staff were sceptical of actually receiving potential benefits.

implicate the failure of Western legal systems, IP law, and ideas about benefit sharing to allow for alternate forms of value, authority, judgment, injury, and remedy in the way knowledge—plant relationships and transactions might be approached. The examples also highlight the common tendency (throughout Thailand) to refer to religiosity, nonconfrontation, and informal measures rather than formal legal ‘rights’ or actions [as discussed in Chiba (1986; 2002), Engel (2001; 2005), and Kasemsup (1986)]. These findings indicate that effective traditional knowledge protection and promotion needs to be reconceptualised—those seeking to engage such communities must respect the diverse and often hybrid customary, value, and authority systems in their engagement with indigenous and local groups.

Issan indigenous seed fair

Finally, a collection of farmers was interviewed and engaged at an ‘indigenous seed fair’. The farmers, who had come mostly from the northeast, but also some from as far as the southern provinces, were exchanging local seeds of plants that could best be described as ‘community knowledge domain’ varieties. It needs to be recognised that this community is not a geographically isolated one, but rather it is a community with “common understandings about the sharing of plants and associated knowledge” (Yuwaa, interview, 2005). These understandings were based on traditions and customs, but were also based on an activism of retraditionalisation. The seed exchange community could best be envisaged as a network of evolving tradition.

Rituals were also explained including the use of various varieties of rice for specific rituals—for example, *khao dam* (or black rice) is highly revered by the local Issan people. A local leader, Ubon Yuwaa (interview, 2005), indicated that rituals and traditional customs in the Issan region were still regularly associated with crops in a similar approach to those expressed by farmers who were interviewed in central Thailand. However, Ubon suggested that these customs are gradually being forgotten throughout Thailand. While showing me parts of the village, Ubon pointed out at nearly every house the presence of shrines dedicated to *Phra-phuum* (spirit lord of the place or village) or *Phii-baan* (the spirit protector of the house), or to Mae Phoesop. These shrines and customs are not limited to the Issan region, but may be found throughout Thailand.

On IP and benefit sharing, local farmers’ groups were hesitantly interested in the prospects of benefits arising through the government’s Plant Variety Protection Fund (see Robinson, 2008a; 2008b). However, they were also sceptical that the government could adequately deliver benefits in a timely, fair, and equitable manner (interviews, Yuwaa, 2005; Siripat, 2005). Thus, it seems that benefit sharing and the various protections of knowledge adopted by the government (under external pressures) offer little more than a distraction from the creation of a ‘universal exchange rate’ for rare and exotic resources to be exploited.

Conclusions

Although the Jasmati trademark is legally valid in the US, to many Thai farmers it is a deceptive name for the product and is therefore considered biopiracy. The transfer of jasmine rice germplasm to the US for plant breeding and induced mutation has been strongly criticised by a range of Thai stakeholders because it disrespects their agricultural heritage, spiritual, and ritualistic links to rice (through rice *kwan* or spirit), was transferred under unclear terms, and may be used to increase the US rice industry’s market share, thus competing against Thai exports. For these reasons, it has been dubbed biopiracy by groups ranging from government officials, academics, NGOs, local industry, and the jasmine rice farmers themselves. The patenting of *kwao krua*

in the US may act as an export concern for similar products. Publicity surrounding the patent has resulted in the overharvesting of the herb. Additionally, certain uses of the plant go against the customary norms of many healers in Thailand. The personification and embodiment of plants and animals, particularly in the context of animist and Buddhist beliefs (such as karma and reincarnation as other beings), also raise concerns about the future potential of biopiracy in the eyes of certain local communities, healers, farmers, and traditionalist parts of the Thai population.

These findings suggest that biopiracy probably cannot be adequately defined through a specific legal or utilitarian mandate, but will need to be contextualised within the geographical and cultural frameworks of many groups and individuals. This study indicates that biopiracy is not just a rhetorical tool for developing countries to seek 'benefits' from the spoils of genetic resource exploitation. It is often rooted in complex cultural and spiritual connections to plants and animals. For an appropriate understanding of these corporeal engagements with animate and vegetative worlds, alternate ways of "seeing, knowing and doing" are required (Howitt, 2001), as well as alternate cartographies and scales that expose the realm of the spiritual (Howitt, 2002), and hybrid notions of nature–society synergies (Whatmore, 1997).

Haraway's ontology of situated knowledges helps lead us in this direction. The situating of discourses and perspectives helps reframe these socioecological projects as relative only to an 'institutionalised' scale (Harvey, 1993, page 41; Whatmore, 1997, page 50), be it global, national, or local. Through heterogeneous actor-networks various groups and individuals with a stake in the issues surrounding biopiracy are attempting to throw off the apparent global hegemony of the IP regime and of the reduction of nature for more nuanced and contextualised understandings.

Last, researchers and policy makers must *not* overgeneralise the unity (or lack thereof) of the many voices concerned about free riding on, and misappropriation of, plants (see Robinson, 2008b). In many articles on bioprospecting and related concepts there has been a tendency for authors to assume—through 'entitlement theory', instrumentalist, utilitarian, and moral rights ideological frameworks—that local plant breeders from indigenous communities and developing countries want compensation and/or exclusive rights of protection (see, for example, Chen, 2006). But these are not universal ideological frameworks and they will not be easily harmonised with the varied traditional laws, customary protocols, and social norms of the diverse local communities in Thailand and many other countries around the globe.

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